Joaquin Auzenne

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Education

University of Texas at Austin — Austin, TX
B.S. Mathematics, B.S. Computational Biology, Certificate in Scientific Computation and Data Sciences — *May 2025*

- College of Natural Sciences Polymathic Scholars Honors
- Thesis: *DNA Metabarcoding of Soil Samples from the Magdalena River Valley*

University of North Texas — Denton, TX
Pre-major studies in Biology and Chemistry (transferred to UT Austin) — *Aug. 2019 – May 2021*

Research Experience

Bioinformatic Assistant — Austin, TX
DiFiore Lab, University of Texas at Austin — *Dec. 2024 – Present*

- Developed a mammalian metabarcoding pipeline for pre-processing and consensus sequence clustering of Nanopore MinION-sequenced environmental DNA (83 soil samples, 6M reads) focusing on rRNA genetic markers (12S, 16S) for longitudinal population and dispersal analyses.
- Integrated multiple public domain databases (NCBI, IUCN) using the GBIF Metabarcoding Data Toolkit API for comprehensive species identification and interactive species-range visualization.

Undergraduate Researcher & Teaching Mentor — Austin, TX
FRI EvoDevOmics, University of Texas at Austin — *Jan. 2022 – Sept. 2023*

- Designed high-throughput transcriptomic pipelines (preprocessing, quality control, alignment) in R on BRCF HPC clusters to reveal the basis of embryonic diversification in Dendrobatid frogs via differential gene expression analyses (DESeq2, WGCNA, PCA, nearest-neighbors clustering).
- Employed Python tool IdMiner for automated literature searches through the PaperBLAST database (PubMed, NCBI) to functionally annotate 40+ differentially expressed genes.
- Served as Teaching Assistant for a computational biology course of 20+ students, presenting weekly lessons on statistical methods (MANOVA, PCA, WGCNA) to student cohorts, omics data analysis, FAIR digital resource stewardship, and career readiness under Rebecca Young, PhD.

Undergraduate Research Assistant — Denton, TX
Henard Lab, University of North Texas BioDiscovery Institute — *Feb. 2020 – May 2021*

- Awarded $500 competitive grant to design a shikimate biosynthesis pathway in plasmid-modified *E. coli* (Gibson assembly) as a "green chemistry" alternative for nylon precursor production in a scalable methane-fed bioreactor.
- Maintained *E. coli* and methanotrophic cultures; evaluated methane production within continuous-stirred tank reactors; performed qRT-PCR confirmation of engineered shikimate-pathway constructs.
- Adapted *"Muconic acid production from methane using rationally-engineered methanotrophic biocatalysts"* (Henard et al.) for *Frontiers for Young Minds*, translating advanced metabolic engineering concepts for a younger audience.

Reptile Care & Research Assistant — Denton, TX
Crossley Lab, University of North Texas Integrative Biology — *Oct. 2019 – Jun. 2020*

- Oversaw daily feeding, tank maintenance, and husbandry for 30+ tanks of large laboratory reptiles (*Alligator mississippiensis*, *Chelydra serpentina*); provided on-call emergency support including animal restraint, transport, and surgical assistance.
- Conducted cardiorespiratory physiological study of alligators and snapping turtles birthed under normoxic and hypoxic conditions via controlled treadmill exercise and blood gas centrifugation.

Professional Experience

Laboratory Operator III — Austin, TX
Emerald Cloud Lab — *Jan. 2026 – Present*

- Execute precision experimental protocols across automated liquid handling, robotic instrumentation, and high-throughput assay platforms in a GMP/GLP-compliant cloud laboratory environment serving multiple simultaneous client projects.
- Operate scientific instrumentation including qPCR thermocyclers, Western Blot, ELISA, HPLC, LC-MS, GC, capillary gel electrophoresis systems for fragment analysis and library QC, flow cytometers, and sequencing platforms.
- Perform routine instrument maintenance, qualification protocols, and sterile-technique procedures (biosafety cabinets, fumehood pipetting, agarose plating, incubation) under standardized operating procedures.

Data Analysis Intern — Dakar, Senegal
Tanel Health (Heman Sweatt Center for Black Males, UT Austin) — *May 2023 – Aug. 2023*

- Implemented gradient boosting and regression (XGBoost, SARIMA) on pharmaceutical sales data to improve back-end inventory forecasting on AWS, with projected waste reduction of up to 20% across 200+ in-network enterprises.
- Created technical reports justifying predictive model selections to shareholders, including associated performance metrics and visualizations (Python: matplotlib, Plotly).
- Digitized prescriptions and integrated a two-factor QR-based prescription authentication system into a 10,000+ customer database, improving compliance and fraud prevention.

Consultant Machinist — Austin, TX
Paradigm Robotics — *Jan. 2024 – Aug. 2024*

- Fabricated mechanical components for fire-resistant robotic systems (FireBot), translating CAD designs into G-code and operating mill/lathe equipment; collaborated with engineers on design iterations and equipment modifications.

CNC Machinist & Programmer — Austin, TX
Carr Lane Manufacturing — *Aug. 2021 – Nov. 2022*

- Trained CNC operators, performed GD&T quality inspections, and maintained 5S standards while machining on HAAS/DMG Mori mills and lathes; programmed overnight production runs increasing weekly machining output by 24+ hours.
- Led evening shift operations, maintaining detailed maintenance logs for tool performance, inventory tracking, and equipment status; documented procedures in SOPs for consistent quality and safety compliance.

Skills

Bioinformatics: Illumina (NextSeq, NovaSeq), ONT (MKICN), QC (Cutadapt, FastQC/MultiQC, NanoPlot), Alignment/Clustering (STAR/Salmon, Kallisto, NGSpeciesID, BWA-MEM2), RNA-Seq (WGCNA, DESeq2), Variant Calling, AlphaFold2
Programming: Python (pandas, scikit-learn, PyTorch, TensorFlow, Biopython), R (tidyverse, ggplot2, Shiny, MCMCglmm, nlme), SQL, Shell Scripting (Bash), MATLAB, High Performance Computing (SLURM), Java, HTML/CSS, C++
Data Analysis: Statistical Modeling (ANCOVA, MLE, PCA, XGBoost, SARIMA), Visualization (ggplot2, matplotlib, Plotly), NLP, GenAI (LLM RAG), Cloud Computing (AWS), FAIR Data Management
Wet Lab: DNA/RNA Extraction, PCR/qPCR/ddPCR, SDS-PAGE, Anaerobic & Microaerophilic Culture (Coy chamber), CRISPR/Cas-Mediated iPSC, Plasmid Design (Gibson Assembly), Light Microscopy, Blood Gas Analysis, GMP/GLP Compliance
Tools & Platforms: LIMS (Benchling), Docker (Bioconductor), Linux, Git/GitHub, VSCode, LaTeX, ArcGIS, MS Office, Adobe Acrobat
Languages: English (Native), French (Fluent), Spanish (Conversational), Japanese (Novice)

Projects

Mammalian Metabarcoding & Species-Range Visualization Platform — Austin, TX
DiFiore Lab, University of Texas at Austin — *Dec. 2024 – Present*

- Developed an interactive web application presenting Nanopore eDNA genomic analyses as user-friendly interfaces for both scientific and public audiences, integrating sequencing results with geospatial species-range data for the Magdalena River Valley.

COVID-19 Spatiotemporal Network Dynamics — Austin, TX
Department of Integrative Biology, University of Texas at Austin — *Oct. 2024 – Dec. 2024*

- Built spatial network time-series models of U.S. COVID-19 spread (32M cases, 04/2020–04/2021) with Shiny interactive maps, dashboards, and statistics to identify transmission hubs from weighted network centrality to inform public-health intervention strategies.

IdMiner — Bioinformatic Literature Search Tool — Austin, TX
FRI EvoDevOmics, University of Texas at Austin — *2022*

- Co-developed a Python-based academic literature search application querying PubMed and NCBI repositories for gene-associated publications to aid researchers in functional annotation of gene lists at scale.

Leadership, Mentorship & Service

EvoDevOmics Peer Mentor (UT Austin FRI, 2022–2023): Mentored 20+ undergraduates in computational biology, omics data analysis, NGS methods, and FAIR data stewardship; fostered independent research skills and career readiness.

Sweatt in Senegal Program (UT Austin, Summer 2023): Selected for inaugural cohort; led cross-cultural data science collaboration in Dakar, Senegal, contributing to healthcare informatics projects in a resource-limited international setting.

Science Communication: Adapted a peer-reviewed metabolic engineering research for *Frontiers for Young Minds*.

Certifications & Training

MSSC Certified Production Technician
Lean Six Sigma White Belt
OSHA-10 General Industry Safety Certification

Publications & Presentations

COVID-19 Spatiotemporal Network Dynamics — Course presentation, Department of Integrative Biology, UT Austin, December 2024.

References

Available upon request. Faculty and supervisor references available from: Dr. Anthony Di Fiore (UT Austin), Dr. Rebecca Young (UT Austin), Dr. Calvin Henard (UNT BioDiscovery Institute).